

Assignment 4: Student Exam Score Prediction (summary)

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dataset	Student Habits vs Academic Performance (kaggle), 1,000 students
my subset	male students: 477. “study hours >= 5” (roll mod 5 + 1) left 80, fewer than 150, so that rule was dropped
cleaning	37 missing parental_education_level values set to “Unknown”, no duplicates, text columns one-hot encoded, gender constant so it drops out
final model	Linear Regression on 7 daily habits: study hours, social media, netflix, attendance, sleep, exercise, mental health

results (80/20 split)

version	test R ²	5-fold CV R ²
v1 study hours only	0.710	0.658
v2 + sleep, attendance, social media	0.752	0.713
v3 + all daily habits	0.916	0.897
v4 + text columns (no gain)	0.913	0.896
v5 weak features dropped (final)	0.916 (train 0.905)	0.898 ± 0.012

feature scaling (v6) gives exactly the same R², as it always does for linear regression. test RMSE 5.17 marks, MAE 4.17. **pass condition (test R² >= 0.70): passed**, no train-test gap.

which habits mattered most

- **study hours:** about +9.6 marks per extra hour a day, by far the strongest (alone it explains about 70%)
- **mental health:** +2.0 marks per rating point, a clear second
- **sleep:** 7 to 8 hours is the best band (74 average, against 65 under 5 hours), +1.9 marks per hour
- **exercise:** +1.5 marks per weekly session
- **screen time:** -2.3 per daily hour of social media, -2.2 per hour of netflix
- **part-time job and attendance:** small (about -2 marks and +0.15 per percent)

why the model performs as it does

the habits hardly move together (no pair above 0.08), so each one explains a different part of the score and a straight line on all 7 fits well. limits: the data looks synthetic and very clean, and scores are capped at 100, which squeezes the top end.